

Writing Large Numbers in Scientific Notation

MA.8.NSO.1.4

I can write a large number in scientific notation, read one back into standard form, and find how many times larger one number is than another.

Yesterday the exponent laws finished — numbers and letters.

Today they go to work on twenty-digit numbers.

Warm-Up

On your own. Four minutes. No notes.

1. Rewrite with a positive exponent: $4x^{-3}$
2. Find the value: $(10^2)^3$
3. Find the value: $\sqrt{121}$
4. In 4,500,000, which place is the 4 in?

Warm-Up

Answers.

1. Rewrite with a positive exponent: $4x^{-3}$ **$4/x^3$**
2. Find the value: $(10^2)^3$ **$10^6 = 1,000,000$**
3. Find the value: $\sqrt{121}$ **11**
4. In 4,500,000, which place is the 4 in? **millions**

Notes

Copy the table.

I. Powers of ten name the places

10^1	10^2	10^3	10^4	10^5	10^6
10	100	1,000	10,000	100,000	1,000,000

The exponent counts the zeros — and names the place of the 1.

Notes

II. Scientific notation

$$a \times 10^n \quad \text{with} \quad 1 \leq a < 10$$

$$2,000,000 = 2 \times 10^6$$

$$26,000,000 = 2.6 \times 10^7$$

The coefficient is at least 1 and less than 10: one nonzero digit before the decimal point.

The exponent is how many places the decimal point moves: 26,000,000. → 2.6, seven places, so 10^7 .

Notes

III. How many times larger?

$$\frac{6.4 \times 10^7}{3.2 \times 10^7} = 2$$

$$\frac{2 \times 10^5}{2 \times 10^2} = 10^3 = 1,000$$

$$\frac{8.4 \times 10^6}{2.1 \times 10^4} = 4 \times 10^2 = 400$$

Divide the coefficients. Divide the powers of 10. Multiply the two results.

On a screen, 8.4E6 means 8.4×10^6 .

Example 1

Standard form to scientific notation.

In 2018, funding for education in Florida totaled 30,100,000,000 dollars.

Express this value in scientific notation.

Example 1

Worked.

$$30,100,000,000 = 3.01 \times 10^{10} \quad \text{the 3 is in the ten-billions place}$$

$$3.01 \times 10,000,000,000 = 30,100,000,000 \quad \text{check}$$

Answer: 3.01×10^{10}

Your Turn

Same steps, your numbers. Boards up when done.

A painting sold for 110,500,000 dollars. Write this value in scientific notation.

Your Turn

Answer.

A painting sold for 110,500,000 dollars. Write this value in scientific notation.

the 1 is in the hundred-millions place

Answer: 1.105×10^8

Example 2

How many times larger?

How many times larger is 8.1×10^8 than 5.4×10^7 ?

Example 2

Worked.

$$\frac{8.1}{5.4} = 1.5 \quad \text{coefficients}$$

$$\frac{10^8}{10^7} = 10^1 = 10 \quad \text{powers of 10: subtract the exponents}$$

$$1.5 \times 10 = 15 \quad \text{multiply the two results}$$

Answer: 15 times larger

Your Turn

Same steps, your numbers. Boards up when done.

Everglades alligators: 2.0×10^5

Florida alligators: 1.5×10^6

About how many times larger is the Florida number?

Your Turn

Answer.

Everglades alligators: 2.0×10^5

Florida alligators: 1.5×10^6

About how many times larger is the Florida number?

$$0.75 \times 10 = 7.5$$

Answer: about 7.5 times larger

Whiteboards • Question 1 of 9

Boards up on three.

Write 4,000,000 in scientific notation.

Answer it.

Which place is the 4 in?

Whiteboards • Question 1 of 9

Answer.

Write 4,000,000 in scientific notation.

4 millions

Answer: 4×10^6

Whiteboards • Question 2 of 9

Boards up on three.

Write 7.2×10^4 in standard form.

Answer it.

Move the point four places.

Whiteboards · Question 2 of 9

Answer.

Write 7.2×10^4 in standard form.

$$7.2 \times 10,000$$

Answer: 72,000

Whiteboards • Question 3 of 9

Boards up on three.

Write 356,000 in scientific notation.

Answer it.

Every nonzero digit stays in the coefficient.

Whiteboards • Question 3 of 9

Answer.

Write 356,000 in scientific notation.

3.56 hundred-thousands

Answer: 3.56×10^5

Whiteboards • Question 4 of 9

Boards up on three.

Which is greater, 9.5×10^5 or 1.2×10^6 ?

Answer it.

Look at the powers first.

Whiteboards • Question 4 of 9

Answer.

Which is greater, 9.5×10^5 or 1.2×10^6 ?

950,000 versus 1,200,000

Answer: 1.2×10^6

Whiteboards • Question 5 of 9

Boards up on three.

Which is 52,000,000 written in scientific notation?

A. 5.2×10^7

B. 52×10^6

C. 5.2×10^6

D. 5.2×10^8

Answer it.

Whiteboards • Question 5 of 9

Answer.

Which is 52,000,000 written in scientific notation?

A. 5.2×10^7

B. 52×10^6

C. 5.2×10^6

D. 5.2×10^8

Answer: A — 5.2×10^7

Whiteboards • Question 6 of 9

Boards up on three.

How many times larger is 6×10^8 than 6×10^5 ?

A. 1,000

B. 3

C. 300

D. 1.6

Answer it.

Whiteboards • Question 6 of 9

Answer.

How many times larger is 6×10^8 than 6×10^5 ?

A. 1,000

B. 3

C. 300

D. 1.6

Answer: A — 1,000

Whiteboards • Question 7 of 9

Boards up on three.

How many times larger is 8.4×10^6 than 2.1×10^4 ?

Answer it.

Coefficients, then powers, then multiply.

Whiteboards • Question 7 of 9

Answer.

How many times larger is 8.4×10^6 than 2.1×10^4 ?

$$4 \times 10^2$$

Answer: 400

Whiteboards • Question 8 of 9

Boards up on three.

A computer displays a number as $3.5\text{E}9$.

Write the number in standard form.

Answer it.

E is not an error.

Whiteboards • Question 8 of 9

Answer.

A computer displays a number as 3.5E9.

Write the number in standard form.

$$3.5 \times 10^9$$

Answer: 3,500,000,000

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Marcus writes 2,000,000 as 20×10^5 .

Is his expression equal to 2,000,000? Is it scientific notation? Explain.

Write your answer in sentences.

Use the definition.

Whiteboards · Question 9 of 9

Answer.

Marcus writes 2,000,000 as 20×10^5 .

Is his expression equal to 2,000,000? Is it scientific notation? Explain.

equal, yes — scientific notation, no

Answer: Equal, yes. Scientific notation, no: 20 is not less than 10. 2×10^6 .

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Stop at a SmartScore of 67 on each skill. Whatever you do not finish is tonight's practice.

Today's skills

Convert between standard and scientific notation: large numbers — 2AL

Compare large numbers — WEZ

Also consider: Powers of ten — LJQ